

Agronomic Introduction

Understanding winter wheat, its cultivation, and its satellite monitoring

EDUCATIONAL DOCUMENT

Master's Thesis

Harvest Window Prediction via Machine Learning

Winter wheat · Centre-Val de Loire · France

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Who is this document for?

This document is deliberately pedagogical. It is intended for anyone who wishes to understand this project without prerequisites in agronomy, remote sensing, or machine learning. You don't need to be an agronomist or engineer to read it — each concept is explained from the beginning, with concrete examples and diagrams.

Why such a document?

Our project combines several technical fields that are not familiar to everyone:

- 🌾 **Agronomy:** understanding how wheat grows, its developmental stages, its needs
- 🛰️ **Remote sensing:** how satellites observe crops from space
- 📡 **SAR radar:** a different technology from optical satellites, which "sees" through clouds
- 🌱 **Soil science:** why soil influences plant growth
- ☀️ **Climatology:** the weather variables that drive growth
- 🤖 **Machine learning:** how models can learn to predict dates

Instead of assuming you know these topics, we introduce them here in a complete and accessible way.

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PART I — Wheat and its cycle

1. What is winter wheat?

1.1 A strategic cereal

Soft wheat (*Triticum aestivum*) is the most cultivated cereal in the world. It is the main ingredient in bread, pasta, pastries, and many other products. In France, wheat represents approximately **5 million hectares** of cultivated area each year — equivalent to half of Belgium's territory entirely dedicated to wheat.

Definition — Soft wheat: species of cereal (*Triticum aestivum*) mainly used for flour, bread, and pastries. It is opposed to "durum wheat" (*Triticum durum*) used for pasta and semolina. In our project, we work exclusively with soft wheat.

1.2 "Winter wheat" — why this name?

There are two major categories of wheat:

Type	Sowing	Harvest	Feature
Winter wheat	October–November	July	Needs cold to flower (vernalization)
Spring wheat	February–March	August	No cold needed, shorter cycle

Winter wheat is largely dominant in France (95% of acreage). It is sown **in autumn**, spends winter as young plants (10-15 cm tall), then resumes its growth in spring to reach 80-110 cm before harvest.

Concrete example: in Centre-Val de Loire (our study area), a typical winter wheat will be sown in late October 2019, will spend winter growing slowly, will produce its ears in spring 2020, and will be harvested in early July 2020. We thus speak of the "2020 season" for this complete cycle.

1.3 Why specifically study wheat?

Winter wheat is particularly interesting for our harvest prediction project for several reasons:

- **Major economic importance:** predicting the optimal date = optimizing millions of hectares
- **Long and well-studied cycle:** ~9 months between sowing and harvest, lots of data available
- **Sensitivity to conditions:** weather, soil, and water stress strongly influence maturity date
- **Effective satellite monitoring:** wheat fields are large and homogeneous, ideal for remote sensing
- **Reference data available:** France has a public service (Céré'Obs) that tracks harvest dates each year

2. The phenological cycle of wheat

Definition — Phenology: study of the calendar of natural phenomena in plants: when they germinate, when they flower, when they ripen. Phenology answers the question “*what happens and when?*” in the life cycle of a plant.

2.1 The major stages of the cycle

From sowing to harvest, a wheat grain goes through **7 main stages**. Here is the journey of a grain sown in October:

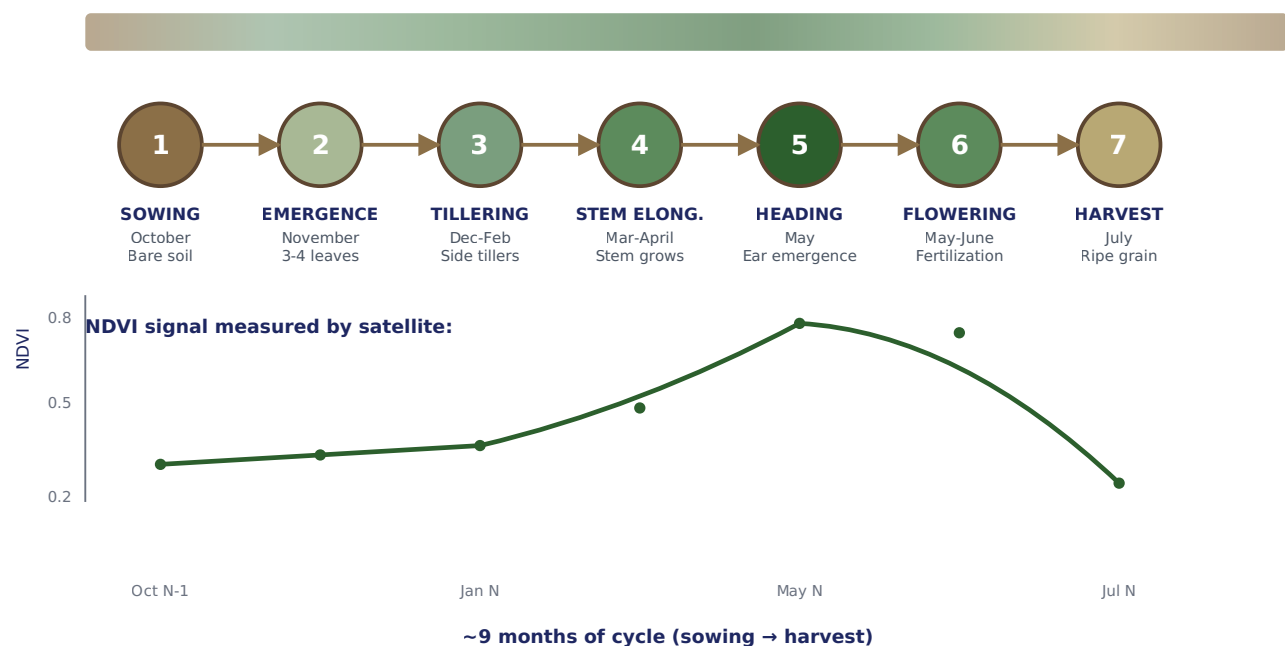


Figure 1.1 — The 7 major stages of the winter wheat cycle, with the typical NDVI signal measured by satellite below. NDVI reaches its maximum at heading-flowering then drops sharply with maturation and harvest.

2.2 Detail of each stage

Stage 1 — Sowing (October)

The wheat grain is sown in the soil tilled by the farmer, at about 2-3 cm deep. At this stage, from space, we only see **bare soil** with a light brown color.

Stage 2 — Emergence (October-November)

The grain germinates and produces its first green leaves (3-4 leaves). The plant measures 5-10 cm. The satellite begins to detect **sparse greenery** in the field.

Stage 3 — Tillering (December-February)

The plant produces “**tillers**”, i.e., side shoots from its base. Instead of having a single stem, wheat will have 3-5 per plant. This is a phase of **slow growth** during winter. The green cover of the field gradually thickens.

Definition — Tiller: secondary shoot that emerges at the base of the plant. Each tiller can produce its own ear. Tillering therefore largely determines the number of ears per square meter — a key factor in final yield.

Stage 4 — Stem elongation (March-April)

The main stem elongates rapidly (1-2 cm per day). The plant goes from 15-20 cm to 60-80 cm in a few weeks. The field becomes **uniformly green and dense**. This is the beginning of the "explosive" phase that leads to ear production.

Stage 5 — Heading (May)

The ear emerges from the sheath of the last leaf and becomes visible. This is a **crucial moment**: the number of grains per ear is now determined. The plant reaches its maximum height (80-110 cm). It is generally at this stage that NDVI reaches its **maximum**.

Stage 6 — Flowering (late May - mid-June)

The tiny flowers inside the ear open briefly (often only 3-5 days) and pollen is released. This is the moment of **fertilization** that will produce the grains. NDVI is still high but starts to decrease.

Stage 7 — Maturation and harvest (June-July)

The grains fill up (the "grain filling" phase), then the plant **senesces** (turns yellow) as the grains harden. When the grain moisture content drops below **14-15%**, the farmer can harvest. The field then goes from green to golden-yellow, and the combine harvester reveals **bare soil**.

Key concept: the moment of harvest is not a fixed date — it depends on grain moisture, which itself depends on weather, soil, and cultivated variety. That's why predicting the "optimal date" is complex and requires lots of data.

2.3 Development speed and "degree-days"

You may have noticed a strange thing: between stage 3 (tillering in December) and stage 4 (stem elongation in March), **3 months** pass. But between stage 4 and 7 (stem elongation to harvest), only **3-4 months** also pass, although the plant undergoes many more transformations. Why?

The answer is in **temperature**. A plant does not grow with passing time, but with **accumulated heat**. In winter, it's cold, and the plant grows very slowly. In spring and summer, heat accumulates, and growth accelerates.

Definition — Growing Degree Day (GDD): measure of heat accumulated by the plant. For wheat, each day we compute: $GDD = (T_{max} + T_{min}) / 2 - T_{base}$, where $T_{base} = 0^{\circ}\text{C}$ (for wheat). We then **accumulate** these values over the entire season. Wheat must accumulate about **2200°C·days** from emergence to harvest.

Numerical example: if a day has a max temperature of 15°C and min of 5°C, the day's GDD is: $(15+5)/2 - 0 = 10^{\circ}\text{C}\cdot\text{days}$. If you accumulate 10°C·days per day for 220 days, you reach 2200°C·days — that's about the wheat maturity threshold. But in reality, we have colder days (winter, ~3°C·days) and hotter days (summer, ~20°C·days), so the actual time varies according to climatic conditions.

2.4 Why is phenology so important?

Understanding wheat stages is essential for **3 reasons**:

1. **Needs change** — Wheat needs more water during stem elongation than at emergence.
2. **Risks change** — Frost in April (during stem elongation) kills young ears, while in December it has no consequences.
3. **Harvest is being prepared** — We only harvest at phase 7. Everything before determines whether we will harvest early (June) or late (August).

2.5 The concept of "optimal harvest window"

When wheat is ripe, the farmer has a **time window** to harvest — not a fixed date. This window typically lasts **7-15 days**, during which:

- The grain has the right moisture (between 12% and 15%)
- The stem is dry enough to be cut cleanly
- Weather conditions are favorable (no rain)
- The farmer has the availability of their combine harvester

Why is this window important?

-  **Harvesting too early:** grain too wet → high drying costs, mold risk
-  **Harvesting too late:** grain too dry → losses during threshing, risk of sprouting on the stalk if rain
-  **Harvesting at the right time:** maximum quality, optimal yield, minimum costs

Our project aims precisely to **predict this optimal window** for each parcel, using satellite, weather, and soil data.

3. The BBCH scale — a universal language

3.1 Why a standardized scale?

Imagine a French farmer says "my wheat is in stem elongation" and a German researcher says "mein Weizen ist in der Schossphase". How to be sure they're talking about the same stage? And how to write computer programs that understand these vague terms?

To solve this problem, the **BBCH scale** was invented. It's a standardized numerical scale, used worldwide, that breaks down the cycle of any crop into **10 main stages** (0 to 9), themselves subdivided into **10 sub-stages** (0 to 9), for a total of 100 possible codes (00 to 99).

Definition — BBCH: German acronym for *Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie* (the three institutions that created this scale in the 1990s). It is today the international standard for describing phenological stages of hundreds of cultivated species.

3.2 The BBCH scale applied to wheat

Here are the main BBCH codes for winter wheat, organized by main stage:

Code	Main stage	Description
0	Germination	From dry grain to coleoptile tip above soil
00		Dry grain
09		Emergence: tip of first leaf visible
1	Leaves	From 1 to 9 leaves unfolded
11		1 leaf unfolded
13		3 leaves unfolded (often ≈ emergence completed)
2	Tillering	Emergence of side tillers
21		1 tiller visible
25		5 tillers (often in winter)
3	Stem elongation	The stem elongates
31		1st node visible (March-April)
37		Last leaf emerging
39		Last leaf fully unfolded
5	Heading	The ear emerges from the leaf sheath
51		Start of ear emergence
55		Half of ear emerged
59		Ear fully emerged
6	Flowering	Anthesis — flowers open
61		Start of flowering
65		Full flowering (50% of flowers open)
69		End of flowering

7	Grain filling	The grain swells and accumulates starch
71		Milky grain (white liquid inside)
75		Medium milk stage
8	Maturation	The grain hardens, plant yellows
83		Soft dough grain
87		Hard dough grain
9	Senescence	Dry plant, ready to harvest
91		Hard grain, plant still green in places
92		Ripe grain, breaks under fingernail → HARVEST ★
99		Post-harvest grain

Concrete example: in the CéréObs public service, the "50%-harvest date" corresponds to BBCH stage 92 reached in **half of the parcels of a region**. It is this date that we use as the target in our ML project.

3.3 The BBCH ↔ NDVI link

NDVI measured by satellite is **not** directly a BBCH stage, but is closely correlated. Here is the typical correspondence:

BBCH stage	Typical NDVI	Interpretation
00–09 (sowing-emergence)	0.15 – 0.30	Dominant bare soil, little vegetation
13–25 (young leaves, tillering)	0.30 – 0.55	Partial green cover
30–39 (stem elongation)	0.55 – 0.75	Dense green cover
50–69 (heading-flowering)	0.75 – 0.90	Season maximum
70–79 (grain filling)	0.65 – 0.80	Still green, but senescence begins
80–89 (maturation)	0.40 – 0.65	Gradual yellowing
90–99 (senescence/harvest)	0.15 – 0.35	Dry plant then bare soil

ML insight: NDVI doesn't say "BBCH 92" but its **sharp drop** from 0.75 to 0.35 indicates the maturation → harvest transition. This is precisely what our model will seek to detect!

4. The French context: Centre-Val de Loire

4.1 Why this region?

Our project focuses on the **Centre-Val de Loire (CVL)** region, which covers central France between Paris and Bordeaux. This region was chosen for **4 reasons**:

1. **Major cereal region:** CVL is one of the largest wheat producers in France, with about 500,000 hectares of wheat each year.
2. **Diverse soils:** the "Beauce" (northeast, deep silty soils) and the "Berry" (south, more clayey soils) offer interesting agronomic diversity.
3. **Temperate climate:** favorable conditions for winter wheat, neither too hot nor too cold.
4. **Reference data available:** Céré'Obs publishes harvest dates for this region each year.

4.2 The 6 departments of CVL

The Centre-Val de Loire region is composed of **6 departments**. Our project covers all six:

Code	Name	Capital	Agronomic feature
28	Eure-et-Loir	Chartres	Heart of Beauce, deep silty soils
45	Loiret	Orléans	Beauce + Loire valley
41	Loir-et-Cher	Blois	Sologne (sandy) + southern Beauce
37	Indre-et-Loire	Tours	Mixed valleys and plateaus
18	Cher	Bourges	Champagne berrichonne, clayey soils
36	Indre	Châteauroux	Boischaut, more constraining soils

4.3 La Beauce — "wheat granary" of France

When we talk about wheat in CVL, we often talk about **la Beauce**. It's a vast sedimentary plain in the northeast of the region (departments 28 and 45 mainly). Its characteristics make it an exceptional cereal terroir:

- **Deep silty soils:** easy to work, good water retention
- **Flat topography:** allows mechanized agriculture on a large scale
- **Temperate oceanic climate:** rainfall distributed throughout the year, summers not too dry
- **Cereal tradition:** lands cultivated for wheat since the Middle Ages

Historical example: as early as the 13th century, Beauce fed Paris with wheat. Today, it produces an average of **7-9 tons of wheat per hectare**, among the best yields worldwide. By comparison, the world average is 3-4 t/ha.

4.4 Our sample: 1500 parcels

For our study, we selected **1500 wheat parcels** in this region:

- **300 parcels per year** (from 2020 to 2024)
- **Minimum area of 1 hectare** (to avoid very small parcels whose satellite signal would be noisy)
- Distributed across the **6 departments**
- Randomly selected (random sampling) from the **RPG** (Graphic Parcel Registry, a public database of all agricultural parcels in France)

Definition — Agricultural parcel: elementary unit of agricultural exploitation. It is a homogeneous land, cultivated in a single manner (one species, one management) by one farmer. A farm can have several parcels, geographically scattered.

PART II — Factors influencing wheat

The development and harvest date of wheat does not depend only on the calendar. Three major families of factors come into play:

1. 🌱 **Soil** (static): composition, structure, water retention capacity
2. ☀️ **Weather** (dynamic): temperature, rainfall, sunshine
3. 🧑 **Agricultural practices** (semi-controlled): sowing date, variety, fertilization

Understanding these factors is essential to understanding **why** we collect the data we collect.

5. Soil and its properties

5.1 Soil is not just a support

For a beginner, "the earth" can seem to be just a place where you plant. In reality, soil is a complex system with its own characteristics, which strongly influence how and when a wheat will develop.

Definition — Soil: living medium composed of mineral particles (sand, silt, clay), organic matter (decomposed plant and animal residues), water, air, and living organisms (bacteria, earthworms, fungi). It is **the interface** between the plant and its mineral environment.

5.2 Texture: sand, silt, clay

Soil is composed of particles of different sizes. According to their proportions, three major **textures** are defined:



Figure 5.1 — The 3 textural fractions of soil. From the largest (sand, visible grains) to the finest (clay, microscopic plates). Their proportions determine how water flows in the soil.

Texture	Composition	Behavior for wheat
Sandy	> 70% sand	Poor soil, water evacuates quickly, wheat suffers in case of drought
Silty	Balanced mix, silt-dominated	★ Ideal for wheat: retains water well, easy to work
Clayey	> 35% clay	Retains lots of water but difficult to work, can suffocate
Silty-clayey	Mix dominated by silt and clay	Very good for wheat (Beauce case)

5.3 Water retention capacity (AWC)

Definition — Available Water Capacity (AWC): amount of water that soil can retain and provide to the plant. It is expressed in **mm of water per meter of soil**. A soil with AWC = 150 mm/m means that in 1 meter of depth, it can provide 150 mm of equivalent rainfall to the plant.

AWC is **crucial** for wheat. If the soil stores a lot of water (deep silty soil), wheat will survive dry periods in May-June. If AWC is low (sandy soil), spring drought can drop yields by 30-50% and bring forward the harvest date by a few days.

Concrete example: in April 2020, a parcel in Beauce (AWC ~200 mm/m) and a parcel in Sologne (AWC ~80 mm/m) receive a week without rain at 25°C. Beauce still has 150 mm of reserve in its soil → wheat continues normally. Sologne only has 30 mm left → wheat begins to stress, its leaves close, its growth slows, and maturation is brought forward by a few days.

5.4 Other soil properties we measure

In our project, we collect **13 soil properties** via the SoilGrids database, measured at **3 depths** (0-5 cm, 5-15 cm, 15-30 cm). Here are the main ones:

Property	Unit	Importance for wheat
Sand, Silt, Clay	% mass	Soil texture — basis of everything
SOC (Soil Organic Carbon)	g/kg	Organic matter = natural fertility
pH (water)	0-14	Acidity; wheat prefers pH 6-7 (slightly acidic)
CEC (Cation Exchange Capacity)	cmol/kg	Soil's ability to retain nutrients
Nitrogen	g/kg	Nitrogen — main nutrient for wheat
Bulk density	kg/dm ³	Soil compactness — affects rooting
Coarse fragments	cm ³ /dm ³	% of stones — limits useful volume

Why 3 depths? Wheat roots descend to 1-1.5 m, but most of the activity happens in the first 30 cm. We therefore sample at 3 levels to capture the zone actually used by the roots.

6. Climate and weather

6.1 The variables that count for wheat

Weather influences wheat development every day. For our project, we collect daily data on **8 main variables**, from which we derive 22 agronomic indicators. Let's see the main ones:

Temperature

This is **the** most important variable. Temperature determines growth rate via degree-days (seen in section 2.3).

- **T_min** (daily minimum): cold night = frost risk
- **T_max** (maximum): warm = rapid development, but > 30°C can stress
- **T_mean** = (T_min + T_max) / 2: basis for computing GDD

Example: in April 2020, CVL experienced an early heatwave (mean T_max 22°C). Cumulative GDD increased faster than normal, which **brought forward the harvest date** by 5-7 days compared to 2021 (cooler April).

Precipitation (rainfall)

Without water, no plant. But water has a dual effect:

Period	Favorable rain	Unfavorable rain
Autumn (sowing-emergence)	Moderate for germination	Excess → impossible sowing
Winter (tillering)	Recharge of soil reserves	—
Spring (stem elongation)	Essential for growth	Drought → yield drops
Summer (maturation)	—	Rain at harvest → wet grain, degraded quality

Sunshine (solar radiation)

Solar radiation (measured in MJ/m²/day) is the **energy** that powers photosynthesis. The more sun, the more the plant produces biomass — provided there is also enough water.

Air humidity and wind

Combined with temperature, they determine how fast water evaporates from soil and plant, which brings us to the next key concept:

6.2 Reference evapotranspiration (ET₀)

Definition — ET₀: reference evapotranspiration, or "evaporative demand" of the atmosphere. It is the amount of water (in mm) that a theoretical well-watered grass crop would evaporate under given conditions. It is, in a way, the atmosphere's "thirst".

ET₀ is computed by a complex formula (Penman-Monteith, FAO-56) that combines temperature, humidity, wind, and radiation. A hot, dry, and windy day will have a high ET₀ (5-7 mm/day), while a cool and humid day will have low ET₀ (1-2 mm/day).

6.3 The water balance

The concept of **water balance** helps determine if a parcel is "thirsty":

Water balance = **Fallen rain** – **ET₀ demand**

If positive: it rained more than the atmosphere "wants" → soil moistens
If negative: it rained less than demand → soil dries out

Example: in May 2020, a parcel receives 30 mm of rain over the month. The cumulative ET0 of the month is 110 mm. Water balance = $30 - 110 = -80 \text{ mm}$. This is very negative → the soil has dried out by 80 mm. If the soil had 150 mm reserve, 70 mm remain → wheat has not (yet) suffered.

6.4 Why is weather not sufficient alone?

Weather strongly influences when wheat will be ripe, but it doesn't say **everything**:

- Two neighboring parcels (same weather) can have different harvest dates depending on their soil
- The cultivated wheat variety also has an impact (early or late)
- Cultural history (rotation, fertilization) influences wheat health

That's why our project combines **weather + soil + satellite observations**: each source brings different and complementary information.

7. Agricultural practices

7.1 What the farmer decides

The farmer makes many choices that influence wheat development. The main ones:

Decision	When?	Impact
Wheat variety	Previous summer	Early or late; stress resistance
Sowing date	October-November	Sowing early → wheat more advanced in spring
Sowing density	Sowing	250-400 grains/m ² depending on context
Fertilization	Winter and spring	Nitrogen, phosphorus, potassium
Phytosanitary treatments	Spring	Disease, pest control
Harvest date	July	Final decision based on grain moisture

7.2 Crop rotation

A parcel does not bear wheat each year. Farmers alternate crops to:

- **Preserve soil fertility** (legumes fix nitrogen)
- **Break the cycle of diseases** specific to wheat
- **Optimize income** (economically interesting rotations)

Typical CVL rotation: Rapeseed → Winter wheat → Winter barley → Peas → Winter wheat. Each parcel only bears wheat about **1 year out of 2-3**. In our project, we verified that wheat persistence from one year to the next is only **17%** — that's why we sample 300 new parcels each year rather than the same 300 parcels 5 times.

7.3 What we DO NOT collect in our project

To remain realistic and reproducible, our project uses **no** individual farmer data:

- ❌ No individual sowing date (private)
- ❌ No cultivated variety (private)
- ❌ No applied fertilization (private)
- ❌ No individual harvest practice (private)

Instead, we use only **open and externally observable data**: satellites (NDVI, SAR), public weather, public soil, and the **regional 50%-harvest date** published by CéréObs.

This is a strength of the project: by using only open data, our method can be **replicated** in any region or country without having to ask each farmer about their practices.

PART III — Satellite remote sensing

This is probably the most exotic part for a beginner: how we use space satellites to monitor wheat growth. We'll break this down into 4 sections.

8. How do satellites "see"?

8.1 The principle of remote sensing

Definition — Remote sensing: set of techniques allowing observation of Earth from space (or an airplane, or a drone) without physical contact with the target. The word comes from the Greek *tele* (at distance) and the Latin *detectio* (action of discovering).

Every object on Earth **sends back** light to space — either by reflecting sunlight (**passive** sensors), or by reflecting a signal sent by the satellite itself (**active** sensors, like radar).

A satellite equipped with a sensor (special camera or radar antenna) measures this reflected light and transforms it into a **digital image**. Depending on the wavelength measured, different information about the surfaces is obtained.

8.2 Wavelengths — beyond what we see

The human eye only sees a **tiny portion** of the electromagnetic spectrum: visible light (red, green, blue). Satellites measure much more than this.

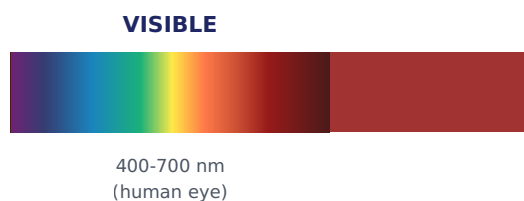


Figure 8.1 — The electromagnetic spectrum and bands used by satellites. The human eye only sees the visible band (400-700 nm). For wheat monitoring, we mainly use near infrared (NIR) and microwaves (radar).

8.3 Why is near infrared (NIR) magical?

Green plants have a very particular property in NIR: they **massively reflect** it. Much more than what they reflect in visible light.

Why? **Chlorophyll** (which gives green color to plants) **absorbs** red photons to do photosynthesis. In NIR, it can't do anything — the cell sends these photons back. The healthier and more photosynthetically active a plant is, the more red it absorbs AND the more NIR it reflects.

Key principle: by comparing what a plant **absorbs** (red) to what it **reflects** (NIR), we have an indirect measure of its photosynthetic activity. This is the basis of NDVI.

8.4 The satellites we use

For our project, we use **two satellite families**, both from the European Copernicus program of the European Space Agency (ESA):

Satellite	Type	Bands	Revisit	Resolution
Sentinel-2	Passive optical	13 bands (visible, NIR, SWIR)	5 days	10-20 m
Sentinel-1	Active SAR radar	C-band (5.4 GHz)	6 days	10 m

These two satellites are **free** and their data has been openly accessible since 2014 (Sentinel-1) and 2015 (Sentinel-2). A revolution for remote sensing research.

9. Vegetation indices (NDVI, EVI, NDWI)

9.1 NDVI — the star index

Definition — NDVI (Normalized Difference Vegetation Index): index that measures the "greenness" and vigor of vegetation. Its formula is:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

NIR = reflectance in near infrared

Red = reflectance in visible red

NDVI varies between **-1 and +1**:

Value	Surface	Interpretation
< 0	Water	Water absorbs both red and NIR
0.0 – 0.15	Bare soil, rocks, sand	No vegetation
0.15 – 0.30	Very sparse vegetation	Soil mostly visible
0.30 – 0.50	Moderate vegetation	Young wheat, stressed grasslands
0.50 – 0.80	Dense growing vegetation	Wheat in full stem elongation
0.80 – 0.95	Vegetation peak	Wheat at heading (max)
0.95 – 1.0	Nearly impossible	Error or extreme case

9.2 Why normalize?

You may wonder: why divide (NIR – Red) by (NIR + Red)? Why not just (NIR – Red)?

The reason: **normalization** cancels out the effects of global illumination. Imagine a parcel photographed at two moments:

- **Moment 1:** sunny sky, lots of light. NIR = 0.6, Red = 0.1.
- **Moment 2:** partly cloudy sky. NIR = 0.3, Red = 0.05.

Without normalization, NIR – Red = 0.5 or 0.25 → two very different values for the **same plant!** With normalization:

- $\text{NDVI}(\text{M1}) = (0.6 - 0.1) / (0.6 + 0.1) = 0.5 / 0.7 = \mathbf{0.71}$
- $\text{NDVI}(\text{M2}) = (0.3 - 0.05) / (0.3 + 0.05) = 0.25 / 0.35 = \mathbf{0.71}$

NDVI is **identical!** NDVI measures an **intrinsic** property of the plant, independent of illumination. This is what makes it robust.

9.3 The typical NDVI profile of wheat over a season

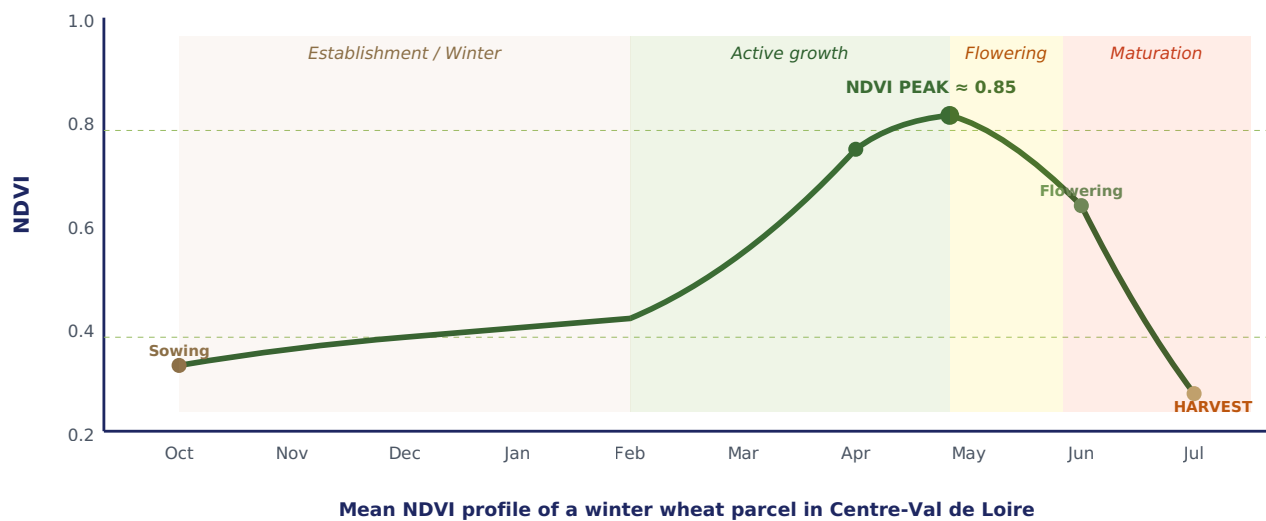


Figure 9.1 — Typical NDVI profile of winter wheat over a complete season. We clearly see the 4 phases: establishment (low NDVI), active growth (rise), peak at heading-flowering, and sharp drop at harvest.

9.4 The other indices: EVI and NDWI

EVI (Enhanced Vegetation Index)

EVI is an **improvement of NDVI**, designed to work better on:

- Very **dense vegetation** (where NDVI saturates around 0.85)
- Visible **light soils** (EVI corrects this effect)
- Atmospheric **residuals**

Its formula, more complex:

$$\text{EVI} = 2.5 \times (\text{NIR} - \text{Red}) / (\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$$

EVI uses **3 bands** instead of 2 (NIR, Red, Blue), and corrects atmospheric effects via the blue band. For our project, this is **complementary information** to NDVI.

NDWI (Normalized Difference Water Index)

NDWI uses SWIR (short-wave infrared) instead of Red:

$$\text{NDWI} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR})$$

SWIR is **absorbed by water**. A well-hydrated plant returns little SWIR (high NDWI), a stressed plant returns more SWIR (low NDWI). NDWI is therefore a **water stress indicator** of the plant.

Example: in June, a parcel has a still high NDVI (0.65) but a falling NDWI (0.10). This indicates a plant that **keeps its green form** but **loses its water** — an early sign of senescence. NDWI can detect stress before NDVI.

9.5 Why combine all three?

Index	Captures	Strength	Weakness
NDVI	General greenness	Very robust, standard	Saturates in dense vegetation
EVI	Vigor (corrected)	Better at peak season	More sensitive to artifacts
NDWI	Leaf hydration	Detects early stress	Noisier than NDVI

Together, these 3 indices give a **more complete view** of plant condition.

10. SAR radar (Sentinel-1)

10.1 A completely different technology

So far, we've talked about **optical** satellites (Sentinel-2) that measure sunlight reflected by Earth. Sentinel-1 uses a **radically different** approach: radar.

Definition — SAR (Synthetic Aperture Radar): radar with synthetic aperture. The satellite **emits** microwave waves toward Earth, and **measures** the echo that returns. It is an **active** sensor (as opposed to passive sensors that just measure received light).

10.2 How SAR radar works

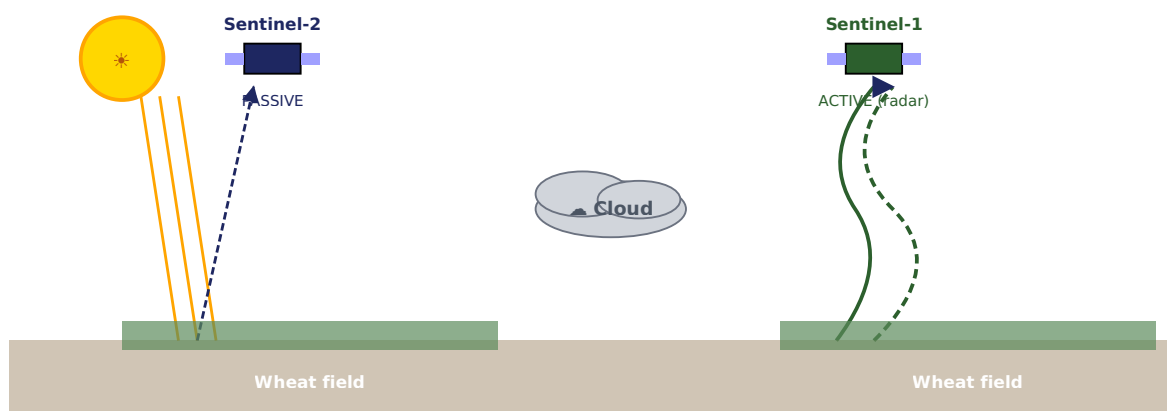


Figure 10.1 — Comparison between Sentinel-2 (passive optical) and Sentinel-1 (active SAR). SAR emits its own wave, which passes through clouds — a major advantage in winter.

The process in 4 steps:

- The satellite **emits** a microwave pulse toward Earth (wavelength ~5.6 cm, C-band)
- The wave travels to the surface in a few milliseconds
- Part of the wave is **sent back** to the satellite (backscatter)
- The satellite **measures** the echo intensity and round-trip time

10.3 Why radar works through clouds

The microwaves used by Sentinel-1 (~5.6 cm wavelength) are **much larger** than water droplets in clouds (~1-100 μm). Consequence: clouds are **transparent** for these microwaves. The radar wave passes through, reaches the soil, comes back, without being disturbed.

The opposite is true for visible light (~500 nm = 0.5 μm), which is **smaller** than droplets — it is scattered by clouds, can't reach the soil, and the optical satellite only sees a white roof.

10.4 What SAR measures: backscatter (σ^0)

Definition — Backscatter σ^0 (sigma zero): backscatter coefficient. It is the percentage of emitted radar energy that returns to the satellite. Expressed in decibels (dB).

For wheat, σ^0 typically varies from **-25 dB to -5 dB**. The higher the value (close to 0), the more the surface returns energy.

What determines the intensity of backscatter?

Factor	Effect on σ^0
Surface roughness	Rougher → more backscatter
Soil moisture	More humid → more backscatter (water reflects)
Vertical structure (canopy)	Vertical stems → strong backscatter in VV
Canopy volume	Dense canopy → attenuates soil signal
Observation geometry	Incidence angle affects echo

10.5 VV and VH polarizations

Sentinel-1 emits and receives in two **polarizations** (wave orientations):

- **VV**: vertical emission, vertical reception (co-polarization)
- **VH**: vertical emission, horizontal reception (cross-polarization)

Why is this interesting? Different structures react differently:

Polarization	Sensitive to	For wheat
VV	Surfaces, vertical stems	Detects bare soil and vertical structure
VH	Volume scattering (canopy)	Detects biomass
VV/VH ratio	Canopy density	Drops when canopy thickens

10.6 The SAR signature of wheat over a season

The SAR profile of wheat is **inverse** to the NDVI profile:

- **Autumn (sowing)**: rough bare soil + debris → **high VV** (~-9 dB)
- **Winter**: low wheat, little canopy → VV stable
- **Spring (stem elongation)**: canopy develops → VV **drops** (attenuation)
- **May (heading-flowering)**: dense canopy → **minimum VV** (~-13.5 dB)
- **June-July (harvest)**: canopy cut → VV **rises sharply**

SAR harvest signal: the **sharp rise** of VV (and the VV/VH ratio) just after harvest is a **direct marker** of the combine harvester passage. This is one of the most valuable signals for our ML model!

11. The cloud problem

11.1 Why clouds are a problem

As we've seen, **optical** satellites (Sentinel-2) cannot "see" through clouds. When a cloud is between the soil and the satellite, the sensor only records the top of the cloud — no information about the field below.

In France and Central Europe, this is a **massive** problem in winter:

Month	% days with dense clouds	Consequence for Sentinel-2
December	50-70%	1 useful image out of 3-4 passes
January	50-70%	Same
February	40-60%	Slightly better
May	20-30%	Good coverage
July	10-15%	Excellent coverage

11.2 How we handle this problem

Our project uses **3 complementary strategies**:

1. **Cloud masking**: we use an algorithm (SCL — Scene Classification Layer) provided by ESA that automatically identifies cloud, shadow, and snow pixels. We **exclude** them before computing indices.
2. **Temporal composites**: we aggregate several scenes over a window (7 days or 15 days) and take the median. If one scene is cloudy but 3 others are clear, the median reflects the clear scenes.
3. **SAR complement**: Sentinel-1 works **all weather**. We thus have ~8 SAR scenes per parcel per month, even in winter. Combined with Sentinel-2, we cover the entire season.

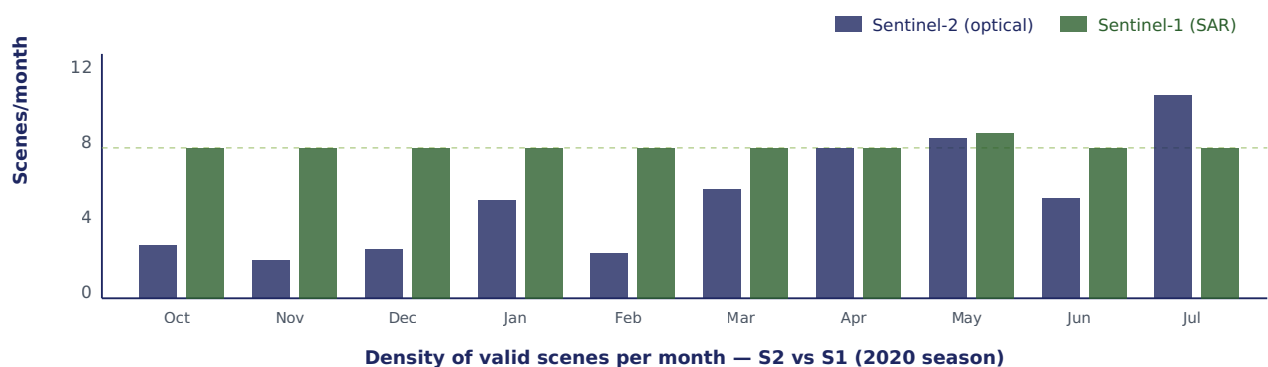


Figure 11.1 — Comparison of valid scene densities between Sentinel-2 (optical, cloud-sensitive) and Sentinel-1 (SAR, all-weather). S1 provides ~8 scenes/month **even in winter**, where S2 drops to 2-3.

Why this combination is unique for our project: few studies combine S1 + S2 at parcel scale over multiple years. It is one of the methodological strengths of our thesis.

PART IV — Harvest and its prediction

12. What is "optimal harvest"?

12.1 The grain must be ripe AND dry

When the farmer considers their wheat ripe, they look at two things:

1. **Physiological maturity:** the grain has reached its final size and weight. This is typically **40-45 days after flowering**.
2. **Grain moisture:** at physiological maturity, the grain still contains ~40% water. We must wait for this water to naturally evaporate to reach 14-15% moisture.

Definition — Grain moisture: percentage of water in the grain relative to its total weight. At harvest, we target **12-15%**. Above → drying costs. Below → risk of breakage at threshing.

12.2 The optimal harvest window

For winter wheat in Centre-Val de Loire, the optimal harvest window is typically located between **early July and mid-July**, but it varies according to:

- The **year** (spring and summer weather)
- The **parcel** (soil, exposure, variety)
- The **region** within CVL

Year	Céré'Obs 50% date	Weather particularity
2020	July 3 (DOY 185)	Hot spring, dry in April-May
2021	July 18 (DOY 199)	Cool spring, 15 days late
2022	July 2 (DOY 183)	Very hot (May heatwave), brought-forward harvest
2023	July 3 (DOY 184)	Average conditions
2024	July 13 (DOY 195)	Wet spring, slight delay

Insight: over 5 years, the date varies by **16 days** (between 2022, the earliest, and 2021, the latest). This is exactly what we want to **predict**: will we be closer to 2022 or 2021 this year?

13. Why predict the harvest date?

13.1 Stakes for the farmer

- **Logistics:** organize their combine harvester, drivers, trailers
- **Cost:** rent the combine at the right week (prices rise in high season)
- **Quality:** harvest at peak quality → better price per kilo
- **Weather risk:** avoid an announced rainy period

13.2 Stakes for the supply chain

- **Cooperatives:** prepare storage silos, plan transport

- **Industrialists:** anticipate raw material arrivals
- **Markets:** the harvest date influences wheat prices
- **National statistics:** track the agricultural campaign in real time

13.3 Scientific stakes

- **Understand climate change:** if harvests advance year on year, it's a signal
- **Improve varieties:** test which varieties adapt to which climate
- **Agricultural policies:** adjust support programs

14. Satellite signals of harvest

How do satellites "see" the harvest?

14.1 NDVI signal: the drop

Before harvest, NDVI gradually goes from 0.7 (green canopy beginning to senesce) to 0.4 (dry plant, still standing) — this is **senescence**.

At the moment of harvest (combine passage), there is a **sharp** break: from 0.4-0.5 (dry plant standing) to 0.15-0.25 (bare soil with some straw). In **less than a week**, NDVI drops 0.25-0.30 points.

ML detection: the **sharp drop of NDVI below 0.4** during the April-July period is a very reliable signal of imminent harvest (within 1-3 weeks).

14.2 SAR signal: the VV/VH ratio rebound

In SAR, it's the opposite:

- **Before harvest:** dense canopy → low VV/VH ratio (~-5-6 dB)
- **During harvest:** canopy disappears, bare soil and straw remain
- **After harvest:** VV/VH ratio rises sharply (~8-10 dB)

SAR therefore detects harvest **even with cloud cover** that would have masked NDVI. This is a **precious redundancy**.

14.3 Weather signal: preparation

Beyond harvest day J, preceding weather variables give clues:

- **Cumulative GDD:** if the crop has accumulated 2200°C·days, it's probably ripe
- **Negative water balance:** senescence accelerates with stress
- **High sunshine in June:** favors rapid maturation

14.4 Combination of all signals

No signal alone is perfect. The ML model will learn to **combine** all these signals to make a robust prediction:

Model logic:

- NDVI dropped below 0.4? Likely imminent harvest.
- But VV/VH ratio not yet recovered? Wait a few days.
- But GDD still low (2050°C·d)? Plant not physiologically mature.
- But soil has low AWC and it's dry? Maturation may accelerate sharply.

Our ML model automatically learns these rules from the 1500 historical parcels.

PART V — The Machine Learning approach

15. Why Machine Learning?

15.1 Limit of traditional approaches

To predict the harvest date, several approaches existed before ML:

Approach	Principle	Limit
Mechanistic agronomic model	Physiological equations (growth, GDD, etc.)	Assumes known variety, ignores actual diversity
Field observations	Parcel visits, manual measurements	Costly, slow, impossible at large scale
Expert rules	"If GDD > X and NDVI < Y then harvest in 2 weeks"	Too rigid, doesn't learn from history

15.2 The advantage of ML

Machine Learning **automatically learns** from historical data. We give it:

- **Examples** (1500 parcels × 5 years, i.e., 1500 cases)
- For each example, **features** (NDVI, SAR, weather, soil) — model input
- For each example, the **true date** of harvest — expected output

The model **learns patterns** that link features and harvest date. Then, given new features (e.g., from 2026), it predicts the expected date.

Definition — Machine Learning (ML): field of artificial intelligence where a program learns to do a task **from data** rather than from explicit rules programmed by a human.

15.3 The types of models we will test

Family	Models	Strength
Decision trees	Random Forest, XGBoost	Robust, interpretable, solid baseline
Recurrent networks	LSTM, GRU	Excellent for time series
Transformer	TFT (Temporal Fusion Transformer)	SOTA for multi-feature temporal
Custom multimodal	Tailored architecture	Truly combines the 6 sources

16. Combined data sources

16.1 Recap of the 6 sources

Source	Role	Type	Volume
Céré'Obs	Target (50%-harvest dates)	1 value / year	5 values
RPG	Parcel identifiers	Static	1500 rows
NASA POWER	Daily weather	Time series	30 features × 1766 days
SoilGrids	Soil (static)	Static	39 features × 1500 parcels

Sentinel-2	Vegetation indices	Time series	15 features × ~118 windows
Sentinel-1 SAR	Radar backscatter	Time series	3 features × ~100 scenes/parcel

16.2 Why combine these 6 sources?

Each brings **unique information** that the others do not:

- **Céré'Obs**: the **ground truth** (without it, no prediction possible)
- **RPG**: the **location** (without it, we don't know what to measure)
- **NASA POWER**: the **climate dynamics** (without it, we ignore accumulated heat)
- **SoilGrids**: the **pedological context** (without it, we ignore why two neighboring parcels differ)
- **Sentinel-2**: the **canopy biophysics** (without it, we don't see the actual plant state)
- **Sentinel-1**: the **structure** (without it, we have no signal in winter)

16.3 Fusion: a technical challenge

Combining 6 sources with **different temporal granularities** (static vs daily vs weekly) is a challenge:

- **Soil** is static (1 value per parcel)
- **Weather** is daily
- **Sentinel-2** is every 5 days (but with cloud gaps)
- **Sentinel-1** is every 6 days (but regular)

Our approach: aggregate all temporal sources into **derived features** per (parcel × year), to produce a single "long" table with 1500 rows and ~200 columns. This will be the **master dataset** we will build at the next stage.

17. Complete glossary

All technical terms used in this document and in the project, classified by theme.

Agronomy and phenology

Term	Definition
Anthesis	Period when flowers open; synonym of flowering. BBCH stage 60-69.
BBCH	International scale (00-99) describing phenological stages. 92 = ripe grain.
Soft wheat	Triticum aestivum species, used for bread, flour, pastries.
Céré'Obs	French public cereal monitoring service (FranceAgriMer + Arvalis).
Heading	Ear emergence from last leaf sheath (May). BBCH 50-59.
Emergence	Appearance of first leaves above soil. BBCH 09-13.
Physiological maturity	State where grain has reached final size and weight (~40 days after flowering).
Stem elongation	Main stem elongation in spring. BBCH 30-39.
Parcel	Elementary unit of agricultural exploitation, homogeneous in crop and management.
Phenology	Study of the calendar of natural phenomena (germination, flowering, etc.).
50%-harvest	Date when 50% of parcels in a region have been harvested.
RPG	Graphic Parcel Registry: French national database of agricultural parcels.
Senescence	Plant aging: leaves yellow, plant dries.
Tiller	Secondary shoot emerging from base of plant. Multiplies number of ears.
Tillering	Phase of tiller emergence (Dec-Feb). BBCH 20-29.
Vernalization	Need for winter cold to trigger flowering (winter wheat characteristic).

Climate and weather

Term	Definition
Water balance	Rainfall – ET0. Positive: soil moistens. Negative: soil dries.
ET0	Reference evapotranspiration, atmospheric water "demand" (mm/day).
FAO-56	Standard method for computing ET0 (Penman-Monteith) published by FAO.
GDD	Growing Degree Days. Heat accumulation measure: $(T_{max} + T_{min})/2 - T_{base}$.
NASA POWER	NASA service providing global weather data at 0.5° resolution.
Penman-Monteith	Complex formula combining T°, humidity, wind, radiation to compute ET0.
Solar radiation	Energy received from sun per unit area (MJ/m²/day).
Water stress	State of plant lacking water for its needs.

Soil science

Term	Definition
Clay	Particles < 0.002 mm. Holds lots of water and nutrients, "heavy" soil.
AWC	Available Water Capacity. Useful water retention capacity (mm/m).
Beauce	Sedimentary plain in NE CVL, foremost French cereal terroir.
CEC	Cation Exchange Capacity. Soil's ability to retain nutrients.
Silt	Particles 0.002-0.05 mm. Ideal texture for wheat (moderate retention).
pH	Soil acidity measure (0=acid, 7=neutral, 14=basic). Wheat: 6-7.
Sand	Particles 0.05-2 mm. Fast drainage, low retention.
SoilGrids	Global soil database at 250 m, provided by ISRIC.
SOC	Soil Organic Carbon. Soil organic carbon = natural fertility.
Texture	Proportions of sand, silt, and clay in soil.

Remote sensing

Term	Definition
Spectral band	Wavelength interval measured by a sensor (e.g., Red = 620-680 nm).
Composite	Image aggregated over a temporal window (e.g., median over 15 days).
Copernicus	European Earth observation space program (ESA).
EVI	Enhanced Vegetation Index. Improved 3-band vegetation index.
GEE	Google Earth Engine. Planetary geospatial analysis cloud platform.
NDVI	Normalized Difference Vegetation Index. $(NIR-Red)/(NIR+Red)$. Greenness.
NDWI	Normalized Difference Water Index. $(NIR-SWIR)/(NIR+SWIR)$. Hydration.
NIR	Near InfraRed (700-1300 nm). Massive reflection by healthy plants.
Pixel	Smallest unit of a satellite image (10×10 m for Sentinel-2).
Reflectance	Fraction of incident light reflected by a surface (between 0 and 1).
SCL	Scene Classification Layer. ESA-provided layer identifying clouds, shadows...
Sentinel-1	Copernicus SAR radar satellites (S1A + S1B), C-band, since 2014.
Sentinel-2	Copernicus optical satellites (S2A + S2B), 13 bands, since 2015.
SWIR	Short-Wave InfraRed (1300-2500 nm). Sensitive to water in vegetation.

SAR Radar

Term	Definition
C-band	Microwave frequency range (~5.4 GHz) used by Sentinel-1.
DESCENDING	Orbit where satellite descends from north to south (morning).

dB (decibel)	Logarithmic unit: $10 \times \log_{10}(\sigma^0)$. Negative values = little backscatter.
GRD	Ground Range Detected. Calibrated and georeferenced Sentinel-1 product.
IW mode	Interferometric Wide. Standard acquisition mode on land, 250 km swath.
Polarization	Orientation of radar wave electric field (V=vertical, H=horizontal).
Backscatter (σ^0)	Fraction of emitted radar energy returning to satellite.
SAR	Synthetic Aperture Radar. High-resolution synthetic aperture radar.
Speckle	Granular noise specific to SAR images. Reduced by spatial filtering.
VH	Vertical emission, Horizontal reception (cross-polarization).
VV	Vertical emission, Vertical reception (co-polarization).



Machine Learning

Term	Definition
Feature	Model input variable (e.g., April mean NDVI, cumulative GDD, etc.).
Feature engineering	Creating new features from raw data (e.g., peak_doy).
GRU	Gated Recurrent Unit. Simplified recurrent neural network.
LSTM	Long Short-Term Memory. Neural network specialized for time series.
Multimodal	Model combining several different data types/sources.
Random Forest	Algorithm based on multiple decision trees, aggregated by voting.
Target	Value to predict by the model (here: harvest DOY).
TFT	Temporal Fusion Transformer. SOTA architecture for time series.
Cross-validation	Evaluation method where part of data is kept for testing.
XGBoost	Very performant and popular tree boosting algorithm.

Conclusion

If you've read this far, well done! 🎉 You now have all the concepts needed to understand what happens in this thesis project.

Recap of key concepts

1. Winter wheat is sown in October, goes through 7 phenological stages (codified BBCH 00-99), and is harvested in July at BBCH stage 92 — when the grain is ripe and dry (12-15% moisture).

2. Development depends on climate (accumulated temperature = GDD, water balance = Rain-ET0) and soil (texture, water retention capacity, fertility).

3. Optical satellites (Sentinel-2) measure canopy greenness (NDVI), vigor (EVI), and hydration (NDWI). But they are blocked by clouds in winter.

4. SAR radar (Sentinel-1) measures canopy structure and volume via backscatter (σ^0) in two polarizations (VV, VH). It works **all weather**, filling winter optical gaps.

5. Our ML approach combines 6 data sources (Céré'Obs target, RPG parcels, NASA POWER weather, SoilGrids soil, Sentinel-2 optical, Sentinel-1 radar) to **learn** to predict harvest date from satellite and climate observations.

What we will do next

Now that data collection is complete, the next steps are:

1. **Assemble the "master dataset"**: merge the 6 sources into a single ML-ready table
2. **Create derived features**: peak_doy, senescence_doy, cumulative GDD, etc.
3. **Train the models**: RF, XGBoost, LSTM, Transformer, multimodal
4. **Evaluate and compare**: which model predicts best? Which features are most important?
5. **Build a dashboard**: visualize predictions, explain results
6. **Write the final thesis**: with all the learnings

And perhaps, at the end, publish a **data paper** in a scientific journal, given the originality of this multi-source combination at the parcel scale in France.

Thank you for reading this document! You are now equipped to understand the rest of the project documentation: the Master Data Dictionary (complete schema of all data) and the Master Dataset Report (complete methodology).